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Price Behavior and Market Integration in European Union Electricity Markets: A VECM Analysis

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Abstract: This study examines the integration and price behavior of European Union electricity markets using a Vector Error Correction Model (VECM). Employing daily wholesale day-ahead electricity prices from 24 EU countries spanning October 2017 to September 2024, the research identifies seven regional clusters of markets based on similarities in price trends. The analysis reveals strong long-term equilibrium relationships and dynamic short-term adjustments, highlighting the interconnectedness of these markets. Central players, such as Germany in Block 1 and France in Block 2, emerge as pivotal in driving regional stability, while markets like Romania and Bulgaria (Block 3) demonstrate significant interconnections. Scandinavian and Baltic regions (Blocks 4 and 5) showcase unique balancing mechanisms influenced by shared infrastructure. Aggregated inter-block dynamics underscore the critical role of central hubs like Blocks 1 and 3 in bridging market disparities. Despite progress, regional heterogeneity persists, with slower adjustments observed in certain clusters. The findings emphasize the need for targeted policies to enhance cross-border electricity trading and infrastructure investments, ensuring equitable integration across all regions. By addressing these disparities, the EU can bolster market efficiency and resilience, contributing to its overarching energy strategy and transition to sustainable energy systems.

Keywords: European electricity markets; market integration; price behavior; cointegration analysis; VECM; regional clustering; energy market dynamics; electricity price convergence

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1. Introduction

The integration of energy markets within the European Union (EU) is a strategic objective aimed at enhancing the efficiency of energy resource allocation, reducing costs for consumers, and increasing resilience to external disruptions, all while facilitating the transition to a sustainable and decarbonized economy [1]. This integration is closely linked to the evolution of electricity markets, where price synchronization and convergence among member states reflect the degree of harmonization of national markets and the efficiency of cross-border mechanisms [2,3]. From this perspective, electricity prices serve as a key indicator of market integration, as they reveal long-term dynamics influenced by fuel prices and provide insights into convergence patterns among European electricity markets through cointegration analysis [4–6].

The integration of EU electricity markets is a complex and multidimensional process, shaped by various factors such as economic, political, and technical ones. Key challenges in this process include rising demand, the variability of renewable energy sources, geopolitical tensions, and the impacts of climate change [2,7,8].

From the perspective of European energy strategy, the primary goal is the establishment of a single energy market. This objective is critical to ensuring affordable and stable energy prices while simultaneously promoting the transition to renewable energy sources.

The history of European energy market integration (Table 1) [9,10] began with market liberalization initiated in the 1990s through the first European legislative packages on energy. However, the process of interconnecting and integrating European energy markets and policies remains ongoing and far from complete.

Table 1. The history of European energy market integration.

Stage Name	Period of Implementation	Description
The First Energy Package	between 1996 and 1998	It introduced the first directives in the energy sector aimed at harmonizing and liberalizing the European Union (EU) internal energy market, creating a more competitive, consumer-centric, and non-discriminatory EU energy market with market-based supply prices.
The Second Energy Package	2003	It allowed industrial and household consumers to select their own energy suppliers from a broader range of competitors.
The Third Energy Package	2009	It established rules for the unbundling of energy supply and production from transmission networks, introduced new requirements for independent regulatory authorities, created a European Agency for the Cooperation of Energy Regulators (ACER), European Network of Transmission System Operators for Electricity (ENTSO-E) and for Gas (ENTSOG), and enhanced consumer rights in retail markets.
The Fourth Energy Package—"Clean Energy for All Europeans"	2019	It introduced new electricity market rules to meet the needs of consumers for secure, sustainable, competitive, and affordable energy while attracting investments, providing consumer incentives, and setting subsidy thresholds to ensure that power plants are eligible to receive subsidies as part of capacity mechanisms. It also required the preparation of risk mitigation plans for electricity crises and enhanced ACER's competences for cross-border cooperation.
The Fifth Energy Package—"Delivering on the European Green Deal" or "Fit For 55"	between 2021 and 2024	Its goal is to align the EU's energy objectives with Europe's new climate ambitions for 2030 and 2050. The package sets a roadmap to achieve a sustainable EU economy by transforming climate and environmental challenges into opportunities, promoting resource efficiency, advancing toward a clean and circular economy, and ensuring a fair and inclusive transition for all.

The liberalization of energy markets opened national markets, introducing competition and fostering the creation of a cross-border energy trading framework [11]. Over time, the objectives have evolved to include carbon emission reductions and the widespread adoption of renewable energy sources. Initiatives such as the European Green Deal underscore the importance of coordination among Member States to achieve climate neutrality by 2050 [12].

The degree of energy market integration can be assessed through price convergence and cointegration. Full integration implies that price variations in one market propagate rapidly and efficiently across all interconnected markets. This is crucial for economic efficiency, as an integrated market allows for optimal resource allocation and reduces unwarranted price fluctuations [4]. However, market integration is not uniform across European regions [13,14]. For instance, some studies reveal that Western European markets exhibit

higher levels of convergence, while Eastern and Southeastern European markets remain relatively isolated due to infrastructural limitations and divergent national policies [15,16].

In this context, analyzing cointegration and the degree of electricity market integration, based on daily prices and the regional clustering of Member States, becomes essential to understanding economic interdependencies and regional dynamics.

Studies investigating energy market integration frequently employ econometric models to evaluate the evolution of short- and long-term interdependencies and the factors influencing them. The Johansen cointegration approach [17] is a foundational model used to test for equilibrium relationships between markets. For instance, the study by Heimeshoff [18] demonstrates that Johansen cointegration analysis is critical for understanding price relationships in European energy markets, highlighting that without addressing capacity constraints, price equality across markets cannot be achieved, thereby underscoring the need for effective market integration strategies. Similarly, Bunn and Gianfreda [19] and Ciferri et al. [6] examined cointegration in European markets, concluding that integration is more advanced among Western countries due to superior infrastructural interconnections.

Volatility-based approaches, such as GARCH models, provide valuable insights into energy market dynamics. These models effectively capture market fluctuations, shedding light on how various factors, including geopolitical tensions and market interdependencies, influence energy prices. Researchers like Brik and Ouakdi [20] demonstrate that geopolitical tensions exacerbate energy market volatility. Recent geopolitical events, including the COVID-19 pandemic and the Russia–Ukraine war, have underscored the interconnectedness of energy and commodity markets, leading to immediate volatility responses [21].

Autoregressive Integrated Moving Average (ARIMA) models have been employed to analyze dynamic relationships between prices and other determinants in energy markets [22]. However, these models primarily focus on short-term relationships and are inadequate for assessing long-term equilibrium between variables [23,24]. In this regard, Vector Autoregressive (VAR) models and Vector Error Correction Models (VECM) have been widely utilized to capture both short- and long-term relationships. Sharma and Mathur [25] emphasize that VECM is ideal for analyzing markets where cointegration relationships are suspected, as it provides insights into long-term equilibrium and short-term adjustments.

The Diebold–Yilmaz model [26–28] uses VAR estimates to compute connectivity indices that highlight the degree of interdependence among variables. The application of the Diebold–Yilmaz model has recently been extended to energy markets [14,29,30]. While this model is innovative and widely used for analyzing market connectivity, it has notable limitations. One significant drawback is its inability to explicitly capture long-term equilibrium relationships between variables, making it more suitable for short-term analysis of volatility and spillovers [26]. Additionally, the model operates within a static framework, rendering it less effective for capturing structural market dynamics, such as exogenous shocks or significant political and economic changes [31]. These limitations make it insufficient for studies where cointegration relationships are essential, such as in interdependent European energy markets.

In this context, the use of Block VECM is becoming increasingly relevant, given their ability to analyze long-term equilibrium relationships and adjustment mechanisms between regional clusters. A Block VECM enables the analysis of relationships not only between individual markets but also among regional clusters, capturing structural interdependencies and providing a more granular understanding of regional dynamics [32]. Moreover, this model can reveal both cointegration relationships and asymmetric behaviors between markets, offering a comprehensive view of short- and long-term connectivity [33]. Thus, while the Diebold–Yilmaz model is effective for analyzing short-term volatility and

shock transmission, studies on European energy markets would benefit more from the application of Block VECM for a thorough understanding of integration mechanisms and interdependencies.

In recent years, machine learning (ML) and artificial intelligence (AI) methods have gained prominence, offering new perspectives on analysis of electricity price dynamics. Techniques such as reinforcement learning, time series generative adversarial networks (GANs), and large language models (LLMs) have been applied to improve forecasting accuracy and capture intricate dependencies. For example, Park and Yang [24] demonstrated the potential of reinforcement learning in dynamic demand management, while Brik and Ouakdi [20] combined GANs with GARCH models to model price volatility under geopolitical pressures. Similarly, LLMs like GPT-4 have shown promise in integrating unstructured data, such as technical reports, news, or research papers, into predictive models [34,35].

Likewise, recent advancements in optimization strategies and technological innovations provide valuable tools for enhancing market integration and sustainability. For instance, Zhong et al. [36] proposed an optimal energy hub operation model that combines distributionally robust optimization methods with Stackelberg game theory, offering solutions for integrating distributed energy resources. Additionally, Lei et al. emphasized the economic and environmental benefits of integrating modern technologies into electricity and carbon markets, while also developing strategies to maximize marginal revenues [37,38].

These models excel at identifying latent patterns and contextual relationships that traditional statistical methods may overlook.

Despite the advancements in AI-based methods, this study prioritizes traditional econometric models, particularly VECM, due to their suitability for structured time series data and interpretability. The structured dataset comprising daily wholesale electricity prices across EU countries aligns well with VECM's capacity to model both short-term and long-term dynamics. Additionally, VECM provides transparent insights into directional relationships and adjustment mechanisms, making it highly relevant for policymakers aiming to enhance market integration and cross-border trade.

Therefore, this research proposes an analysis of European markets through the application of the VECM model, focusing on relationships between regional clusters. The use of daily data enables a detailed capture of market dynamics, while the regional cluster analysis highlights market heterogeneity. This approach combines advanced econometric methodology with an innovative regional perspective, contributing significantly to the existing body of literature.

2. Methodology and Data Collection

The methodology is structured into four distinct steps that progressively build on each other, ensuring a thorough analysis of market relationships, integration, and adjustment mechanisms. Each step offers a critical layer of insight, beginning with the identification of similar behavior across markets, moving on to verifying long-term equilibrium relationships, and concluding with an examination of the dynamic interactions within these clusters (blocks).

2.1. Dynamic Time Warping (DTW)

The clustering of European electricity markets was conducted using the Dynamic Time Warping (DTW) clustering algorithm, a robust method designed to measure similarities between time series data while accommodating temporal misalignments. This method was

selected due to its ability to capture complex variations in electricity price trends across different countries, including seasonal patterns and lagged effects.

The clustering process began by calculating pairwise dissimilarities between the daily electricity price time series for all countries in the dataset. The DTW algorithm quantified the differences between each country's price dynamics, creating a comprehensive cost matrix.

The cost matrix was used as an input for hierarchical clustering. This method enabled the identification of groups of countries with similar electricity price behaviors, reflecting the structural and economic interdependencies within European electricity markets.

The *DTW* algorithm seeks an optimal alignment path between the two series such that the sum of distances along the alignment trajectory is minimized. The DTW distance between two time series, X and Y, is defined as follows:

$$DTW(X,Y) = min_{\phi} \sum_{i,j \in \phi} d\left(x_i, y_j\right) \quad (1)$$

where ϕ represents an alignment path that links each element of series X with one or more elements of series Y, and $d\left(x_i y_j\right)$ is the Euclidean distance between values x_i and y_j:

$$d(x_i, y_i) = \left(x_i - y_j\right)^2 \quad (2)$$

To determine the optimal path, DTW uses a cost matrix D, where each element $D(i,j)$ represents the minimum cost of aligning elements x_i with elements y_j. The cost matrix is filled using the recurrence relation:

$$D(i,j) = d\left(x_i, y_j\right) - min\{D(i-1,j), D(i,j-1), D(i-1,j-1\,\} \quad (3)$$

After calculating the *DTW* distances between all pairs of countries, we constructed a distance matrix that was used to apply a clustering method. We used hierarchical agglomerative clustering to group countries with similar electricity price trends. This method starts by considering each country as a separate cluster, then iteratively combines clusters based on the *DTW* distances between them until all countries are grouped into clusters.

$$D(A,B) = max\{d(x,y) : x \epsilon A, y \epsilon B\} \quad (4)$$

To determine the optimal number of clusters, we used the Silhouette method, which evaluates how well each country fits into its cluster compared to other clusters.

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (5)$$

where

- $s(i)$ is the Silhouette score for point i;
- $a(i)$ is the average distance between point i and all other points in the same cluster (cohesiveness);
- $b(i)$ is the smallest average distance between point i and all points in any other cluster (separation).

The Silhouette Score is a widely used metric for assessing the quality of a clustering solution. The Silhouette Score offers several advantages when evaluating the quality of clustering solutions. By calculating the balance between intra-cluster cohesion and inter-cluster separation, the Silhouette Score ensures that the clustering structure is meaningful and well-defined. Additionally, it is highly useful for determining the optimal number of clusters, as the average Silhouette Score across all data points can guide the selection process,

helping to identify the configuration that best captures the underlying data structure. Another significant advantage of the Silhouette Score is its interpretability. It highlights potential issues, such as overlapping or poorly defined clusters, and offers insights into the overall clustering performance. The metric is also highly flexible, as it can be applied to any clustering algorithm and distance metric, making it suitable for a wide range of datasets and domains.

To ensure meaningful and economically relevant groupings, specific adjustments were made to the cluster composition. Clusters that initially consisted of only one country were merged with adjacent clusters that exhibited geographical proximity and similar market characteristics. Additionally, redundancies within clusters were carefully addressed by evaluating highly correlated markets. This step ensured that the analysis remained statistically robust and avoided issues of multicollinearity.

2.2. Johansen Cointegration Test

The second step involved applying the Johansen cointegration test to verify the cointegration relationships within each cluster (block) of countries. We determined the number of cointegration vectors using two types of statistics: trace statistic and maximum eigenvalue statistic.

(a) Johansen trace test

The trace test (λ_{trace}) evaluates the null hypothesis that there are at most r cointegration relationships against the alternative that there are more than r. The test statistic is given by:

$$\lambda_{trace} = -T\sum_{i=r+1}^{k} \ln(1-\lambda_i) \quad (6)$$

where T is the number of observations, λ_i are the eigenvalues, and k is the number of variables in the system.

(b) Johansen maximum eigenvalue test

The maximum eigenvalue test (λ_{max}) evaluates the null hypothesis of r cointegrating vectors against the alternative of $r+1$ cointegrating vectors. The test statistic is given by:

$$\lambda_{max} = -T\ln(1-\lambda_{r+1}) \quad (7)$$

The results of the Johansen test provide information on the number of cointegration relationships, which implies how many long-term equilibrium relationships exist among the time series in each cluster.

2.3. Vector Error Correction Model (VECM)

The third step involved applying the Vector Error Correction Model (Block VECM) to analyze the dynamic relationships within each cluster (block) of countries, which were determined based on the clustering results from the first step. VECM is suitable for capturing both short-term and long-term relationships among time series that are cointegrated.

The VECM can be used when the variables are non-stationary but cointegrated. It combines short-term dynamics with the long-term equilibrium relationships identified through cointegration analysis. It allows for a more comprehensive understanding of how deviations from the long-term equilibrium impact the short-term behavior of the variables.

For each block (cluster) of countries, the VECM can be represented by the following equation:

$$\Delta Y_t = \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \mu + \epsilon_t \quad (8)$$

where

- ΔY_t represents the vector of first differences in the time series at time t;

- ΠY_{t-1} is the error correction term, where Π captures the long-term equilibrium relationships between the variables;
- $\Gamma_i \Delta Y_{t-i}$ represents the short-term dynamics, with Γ_i being the short-term adjustment coefficients;
- μ is the vector of constants (intercepts);
- ϵ_t is the vector of error terms (shocks).

The matrix Π can be decomposed as follows:

$$\Pi = \alpha \beta' \tag{9}$$

where α represents the speed of adjustment coefficients, indicating how quickly the system returns to equilibrium after a deviation, and β contains the cointegrating vectors that describe the long-term relationships between the variables.

2.4. Block VECM Between Clusters/Blocks

The fourth step involved applying the Block Vector Error Correction Model (Block VECM) between different clusters (blocks) to analyze the relationships and dynamic interactions between the blocks of countries identified in the previous steps. Each block is represented by a time series, calculated as the weighted average (based on the national available electricity for final consumption) electricity price of the countries within that block.

The Block VECM can be expressed by the following generalized form:

$$\Delta Z_t = \Pi Z_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Z_{t-i} + \mu + \epsilon_t \tag{10}$$

where

- ΔZ_t represents the vector of first differences in the block-level time series at time t;
- ΠZ_{t-1} is the error correction term, where Π captures the long-term equilibrium relationships between the blocks;
- $\Gamma_i \Delta Z_{t-i}$ represents the short-term dynamics between the blocks, with Γ_i being the short-term adjustment coefficients;
- μ is the vector of constants (intercepts);
- ϵ_t is the vector of error terms (shocks).

The error correction matrix Π can be decomposed as follows:

$$\Pi = \alpha \beta' \tag{11}$$

where α represents the speed of adjustment coefficients, which indicate how quickly each block returns to equilibrium after a deviation, and β contains the cointegrating vectors, which describe the long-term relationships between the blocks.

2.5. Data Description

We conducted our analysis based on a dataset of daily wholesale day-ahead electricity prices for 24 European Union countries, covering the period October 2017–September 2024. Three countries were excluded from our study (Cyprus, Ireland and Malta), firstly due to data unavailability. Secondly, the exclusion of Cyprus, Malta, and Ireland from the analysis is rooted in the distinct characteristics of their electricity markets, which differ significantly from those of continental European countries. These three countries are geographically isolated as islands, which has a direct impact on their electricity price formation mechanisms. Unlike the interconnected electricity markets on the European continent, island nations often rely on limited domestic generation capacity and lack

access to large-scale, cross-border electricity trading. Cyprus and Malta depend heavily on imported fossil fuels for electricity generation, leading to price volatility closely tied to international fuel markets rather than regional electricity market dynamics [39]. Moreover, their limited grid infrastructure and lack of interconnections with other countries result in less competitive market structures, further distinguishing their price-setting processes. As per the European Network of Transmission System Operators for Electricity [2], Cyprus and Malta remain unconnected to the European electricity grid, preventing participation in the integrated European market. Ireland, while somewhat interconnected with the United Kingdom, still operates under different market conditions compared to continental Europe. Its Single Electricity Market (SEM), shared with Northern Ireland, features unique regulatory and market mechanisms tailored to its specific context as an island system, including capacity remuneration arrangements to ensure supply security [40].

Data were extracted from the Ember online platform (https://ember-energy.org/, accessed on 16 November 2024). We also used a dataset of the annual quantity of electricity available for final consumption, extracted from Eurostat.

3. Results and Discussion

3.1. Dynamic Time Warping (DTW) Clustering Approach

The results of the Dynamic Time Warping (DTW) clustering analysis provided a clear segmentation of the European electricity markets based on the similarity of their price patterns.

The Silhouette analysis further validated the clustering results by determining the optimal number of clusters (Figure 1).

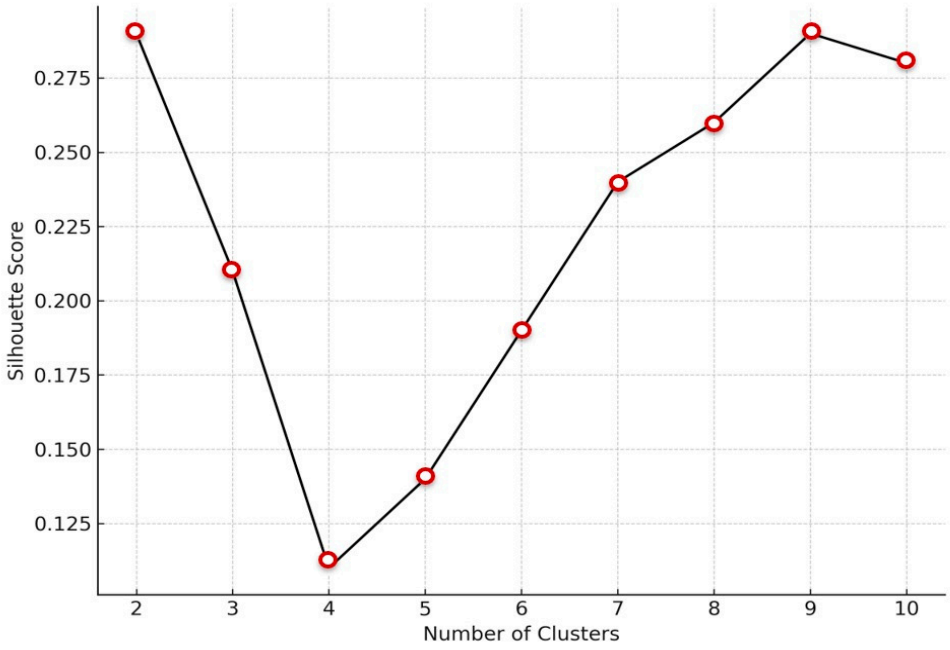


Figure 1. Silhouette scores.

The analysis indicated that either two or nine clusters were optimal, both yielding comparable Silhouette scores. However, the choice of nine clusters was preferred as it provided greater insight into the distinct characteristics of each group, thus supporting a more targeted analysis of market interdependencies and regional variations.

Table 2 presents the composition of the nine clusters, detailing the countries included in each group.

Table 2. Cluster composition of the European electricity markets.

Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7	Cluster 8	Cluster 9
Austria Czechia Germany Luxembourg Slovakia	Belgium France Netherlands	Bulgaria Croatia Hungary Romania Slovenia	Denmark	Estonia Latvia Lithuania	Finland	Greece Italy Poland	Portugal Spain	Sweden

3.2. Johansen Cointegration Test

Before performing the Johansen cointegration test for each cluster (block), some adjustments on the structure of the clusters were implemented.

In the first step, we merged the initial single-country clusters. The decision to merge the initial single-country clusters (clusters 4, 6, and 9) into a single cluster, designated as Cluster 4, was guided by several key factors that reflect the economic, geographical, and structural similarities of the included countries—Denmark, Finland, and Sweden.

Firstly, the geographical proximity of these countries plays a crucial role. Denmark, Finland, and Sweden are all located in the Nordic region, forming a tightly integrated area in terms of energy infrastructure and market operations. Their electricity markets are connected through the Nord Pool power market, one of the oldest and most advanced electricity markets globally [41]. This high level of interconnection ensures a significant degree of price convergence and market synchronization across these countries, justifying their analysis as a single entity.

Secondly, these countries share a similar energy generation profile, with a particularly high reliance on renewable energy sources. According to the European Commission and Nordic Energy Research, electricity generation in Denmark, Finland, and Sweden is predominantly based on wind, hydro, and nuclear energy, with renewables accounting for a substantial portion of their energy mix [39]. This similarity in generation profiles further supports the rationale for grouping these countries, as their electricity prices are influenced by comparable supply-side dynamics.

Thirdly, these markets exhibit strong policy and regulatory coordination, further enhancing their integration. The Nordic countries have established shared energy market policies and regulations that facilitate cross-border electricity trade and ensure the harmonization of market operations [42]. This policy alignment ensures that external shocks or supply–demand dynamics in one country often have spillover effects on its neighbors.

Finally, from a methodological standpoint, the merger was necessary for the application of the Vector Error Correction Model (VECM), which requires at least two countries for meaningful analysis. Grouping these closely related markets not only ensured methodological feasibility but also enhanced the robustness of the results by capturing the collective dynamics of this highly integrated region.

In the second step, Luxembourg was excluded from Cluster 1 due to its exceptionally high correlation with Germany, which was also part of the same cluster. Specifically, the correlation coefficient between Luxembourg and Germany's electricity prices was +1, indicating that their price movements were nearly identical over the analyzed period. This high degree of correlation reflects Luxembourg's reliance on imports from Germany to meet its electricity demand, with Germany acting as a dominant supplier in the region. According to Eurostat, Luxembourg imports over 85% of its electricity, primarily from Germany, which explains the perfect synchronization of their price dynamics [43].

Including Luxembourg in the analysis alongside Germany would have introduced redundancy into the cluster composition. Such redundancy could compromise the statistical rigor of the Johansen cointegration test, as multicollinearity among variables can

distort cointegration relationships and lead to unstable model estimations. By excluding Luxembourg, we ensured that the remaining countries in Cluster 1 represented distinct price-setting dynamics while preserving the statistical integrity of the analysis.

Moreover, this exclusion aligns with the study's broader goal of capturing meaningful differences and interdependencies between countries within each cluster. Given Luxembourg's heavy dependence on German electricity imports and its negligible role as an independent price-setter, its exclusion allowed for a more balanced and representative analysis of the cluster's dynamics. These adjustments provided a more suitable foundation for the application of the Vector Error Correction Model (VECM), as the analysis focused on countries with independent and non-redundant market behaviors.

The methodological adjustments were carefully considered to enhance the robustness and interpretability of the results. These decisions contributed to the robustness of the results by: (a) aligning the clusters with the economic realities of interconnected electricity markets, thereby ensuring that the analytical framework accurately reflects market dynamics; (b) mitigating statistical issues such as multicollinearity and redundancy, which could distort the cointegration and VECM results; (c) strengthening the interpretability of findings by grouping countries with meaningful economic connections and excluding entities with negligible or redundant contributions to price dynamics.

Table 3 shows the final composition of the clusters (blocks), detailing the countries included in each group.

Table 3. Final cluster (block) composition of the European electricity markets.

Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Block 7
Austria Czechia Germany Slovakia	Belgium France Netherlands	Bulgaria Croatia Hungary Romania Slovenia	Denmark Finland Sweden	Estonia Latvia Lithuania	Greece Italy Poland	Portugal Spain

The results for Johansen trace test and Johansen maximum eigenvalue test consistently indicated multiple cointegrating equations (equal to the number countries from each block), suggesting strong long-term linkages between the countries in each block. The electricity prices in each of the seven blocks are highly integrated and have well-established long-term equilibrium relationships. Moreover, the results showed multiple cointegrating equations for all seven blocks (equal to the number of blocks).

3.3. Vector Error Correction Model (VECM)

3.3.1. Vector Error Correction Model (VECM) for Block 1

The VECM results for Block 1, among electricity prices in Austria, Czechia, Germany, and Slovakia, are revealed in Tables 4 and 5.

Table 4. Cointegration equations for Block 1.

	CointEq1	CointEq2	CointEq3	CointEq4
Austria (−1)	1.000	−0.423 (0.029)	1.038 (0.072)	3.054 (0.218)
Czechia (−1)	−2.367 (0.082)	1.000	−2.457 (0.088)	−7.229 (0.227)
Germany (−1)	0.963 (0.045)	−0.407 (0.019)	1.000	2.942 (0.146)
Slovakia (−1)	0.328 (0.067)	−0.138 (0.024)	0.340 (0.072)	1.000
C	0.378	−0.160	0.392	1.154

Table 5. Error correction mechanism for Block 1.

	D (Austria)	D (Czechia)	D (Germany)	D (Slovakia)
Error Correction Term	−0.191 ***	0.095 ***	−0.365 ***	−0.017
	(0.029)	(0.029)	(0.038)	(0.026)
D (Austria (−1))	−0.278 ***	0.034	0.237 ***	0.046
	(0.046)	(0.047)	(0.06)	(0.042)
D (Czechia (−1))	0.009	−0.102 *	−0.208 ***	0.170 ***
	(0.061)	(0.061)	(0.080)	(0.055)
D (Germany (−1))	0.103 ***	−0.036	−0.187 ***	0.030
	(0.032)	(0.033)	(0.043)	(0.030)
D (Slovakia (−1))	−0.089 *	−0.138 ***	−0.104 *	−0.474 ***
	(0.048)	(0.048)	(0.063)	(0.043)
C	0.001	0.001	0.001	0.001
	(0.009)	(0.009)	(0.012)	(0.008)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The long-term dynamics, as captured by the cointegrating equations, highlight Czechia's strong positive influence (+2.367) on Austria, indicating that price increases in Czechia correspond to increases in Austria over the long term. In contrast, Germany (−0.963) and Slovakia (−0.328) exert negative long-term effects on Austria, suggesting balancing interactions where price increases in Germany and Slovakia are associated with reductions in Austrian prices. Czechia itself is positively influenced by Austria, Germany, and Slovakia in the long term, reflecting a mutual alignment within the block. Germany, as normalized in its respective equation, experiences strong positive influence from Czechia (+2.457), while Austria (−1.038) and Slovakia (−0.340) negatively affect it, indicating competitive or stabilizing dynamics. For Slovakia, Czechia (+7.229) emerges as the dominant influencer, followed by Austria and Germany, both of which exhibit negative impacts.

Short-term adjustments show that Germany plays the most significant role in stabilizing the block, as indicated by its strong and rapid error correction term (0.365). Austria also adjusts to long-term imbalances (−0.191) but at a slower rate, while Czechia adjusts positively (0.095), albeit less prominently. Slovakia demonstrates a negligible adjustment (−0.017), underscoring its passive role in maintaining long-term equilibrium.

The short-term dynamics further underscore Germany's centrality, as its price changes significantly influence Austria (0.103). Austria's price adjustments exhibit a self-correcting mechanism (−0.278) and a notable positive influence on Germany (0.237), reinforcing their bilateral importance. Czechia exerts a positive influence on Slovakia (0.170), showcasing regional interconnections, while Slovakia exhibits strong self-regulation (−0.474).

Block 1 exhibits rapid adjustment dynamics, as indicated by the statistically significant and relatively high coefficients of the error correction terms. This behavior can be attributed to the region's well-developed and highly interconnected electricity infrastructure, exemplified by the Central European synchronous grid. Germany, being the largest electricity market in Europe and a major exporter of energy to neighboring countries, plays a pivotal role in stabilizing regional price dynamics through its extensive cross-border interconnection capacities. Furthermore, the integration of renewable energy sources in Germany, such as wind and solar power, and the corresponding market mechanisms (e.g., real-time balancing and intraday trading) enable quick adjustments to price imbalances. Additionally, coordinated market policies within this block, particularly under the framework of the European Union's Internal Electricity Market (IEM), facilitate efficient price convergence. The participation of all four countries in the common bidding zones and regional capacity markets enhances market transparency and ensures that deviations from equilibrium are corrected swiftly.

Overall, Block 1 demonstrates high integration, with Germany acting as the primary driver of market dynamics. Austria aligns closely with Germany, reflecting their strong interdependence, while Czechia serves as a bridge within the block, influencing Slovakia and balancing Germany. Slovakia's role, though less prominent, remains essential in maintaining regional stability.

3.3.2. Vector Error Correction Model (VECM) for Block 2

The VECM results for Block 2, illustrating the dynamics among electricity prices in Belgium, France, and the Netherlands, are highlighted in Tables 6 and 7.

Table 6. Cointegration equations for Block 2.

	CointEq1	**CointEq2**	**CointEq3**
Belgium (−1)	1.000	−3.660 (0.114)	−1.374 (0.036)
France (−1)	−0.273 (0.020)	1.000	0.375 (0.033)
The Netherlands (−1)	−0.728 (0.023)	2.664 (0.123)	1.000
C	0.017	−0.062	−0.023

Table 7. Error correction mechanism for Block 2.

	D (Belgium)	**D (France)**	**D (The Netherlands)**
Error Correction Term	−0.875 *** (0.051)	−0.390 *** (0.050)	−0.200 *** (0.042)
D (Belgium (−1))	0.023 (0.045)	0.234 *** (0.043)	0.089 ** (0.037)
D (France (−1))	0.078 ** (0.037)	−0.248 *** (0.035)	0.156 *** (0.030)
D (The Netherlands (−1))	−0.305 *** (0.046)	−0.200 *** (0.043)	−0.500 *** (0.038)
C	−0.001 (0.009)	−0.001 (0.008)	0.001 (0.007)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The cointegrating equations reveal that, in the long term, Belgium's prices are positively influenced by France (+0.273) and The Netherlands (+0.728), indicating that price increases in these countries drive corresponding increases in Belgium. France's prices, in turn, are positively influenced by Belgium (+3.660) and negatively influenced by The Netherlands (−2.664). For the Netherlands, both Belgium (+1.374) and France (−0.375) exhibit mixed effects, with Belgium exerting a stronger positive influence on the Dutch market compared to France's weaker negative influence.

The error correction terms suggest that all three countries adjust significantly to long-term disequilibria. Belgium shows the fastest adjustment (−0.875), followed by France (−0.390) and The Netherlands (−0.200), underscoring their interconnected nature and alignment towards equilibrium.

In terms of short-term dynamics, The Netherlands has the strongest self-correcting mechanism (−0.500), suggesting robust internal stabilization. Belgium is positively influenced by its own past values (0.023) and by France (0.078), while The Netherlands has a significant negative effect on Belgium (−0.305), reflecting balancing dynamics. France's price movements are positively influenced by Belgium (0.234) and negatively influenced

by The Netherlands (−0.200), while being negatively impacted by its own lagged values (−0.248), indicating both self-correction and mixed external influences. The Netherlands, while self-regulating, is positively affected by Belgium (0.089), but negatively impacted by France (−0.200).

Block 2 displays moderate adjustment dynamics. These trends can be attributed to the region's established electricity markets and infrastructure. France, as a major electricity producer with a significant share of nuclear power, provides a stable and low-cost energy supply that influences neighboring markets. However, the relatively inflexible nature of nuclear energy, combined with intermittent renewable energy integration, can lead to slower responses to short-term price shocks. Belgium and the Netherlands, on the other hand, are characterized by extensive cross-border electricity trading facilitated by strong interconnection capacities. The Netherlands acts as a transit hub for electricity flows between Northern and Western Europe, benefiting from its access to diverse energy sources, including natural gas. Belgium's reliance on imported electricity and its energy-intensive industrial sector creates price sensitivity to changes in supply and demand, which also influences the dynamics within the block. Additionally, the market mechanisms within these countries, including participation in the EU's Internal Electricity Market (IEM), ensure price convergence over time.

Overall, Block 2 demonstrates strong integration, with Belgium and France playing central roles in driving the block's dynamics. Belgium adjusts most quickly to deviations from equilibrium, while The Netherlands exhibits strong self-correction. France acts as a bridge, influencing and being influenced by both neighboring markets.

3.3.3. Vector Error Correction Model (VECM) for Block 3

The VECM results for Block 3, comprising Bulgaria, Croatia, Hungary, Romania, and Slovenia, are presented in Tables 8 and 9.

Table 8. Cointegration equations for Block 3.

	CointEq1	**CointEq2**	**CointEq3**	**CointEq4**	**CointEq5**
Bulgaria (−1)	1.000	−24.098 (0.312)	29.467 (0.381)	−0.975 (0.012)	45.565 (0.589)
Croatia (−1)	−0.041 (0.039)	1.000	−1.228 (1.077)	0.040 (0.038)	−1.891 (1.515)
Hungary (−1)	0.034 (0.053)	−0.818 (1.195)	1.000	−0.033 (0.040)	1.546 (2.115)
Romania (−1)	−1.025 (0.030)	24.706 (0.760)	−30.210 (0.728)	1.000	−46.715 (1.424)
Slovenia (−1)	0.022 (0.040)	−0.529 (0.818)	0.647 (1.029)	−0.021 (0.038)	1.000
C	0.088	−2.125	2.599	−0.086	4.018

The cointegrating equations highlight significant long-term dependencies, with Romania and Bulgaria playing central roles. Romania strongly influences Bulgaria positively in the long term, while Hungary and Slovenia exhibit smaller, balancing effects. Conversely, Bulgaria has a reciprocal influence on Romania, indicating mutual dependence. Croatia, though less central, shows connections with Romania and Slovenia, reflecting the integrated nature of these markets. Hungary is notably impacted by Romania in the long term, with strong positive influences suggesting competitive or balancing interactions.

Table 9. Error correction mechanism for Block 3.

	D (Bulgaria)	D (Croatia)	D (Hungary)	D (Romania)	D (Slovenia)
Error Correction Term	−0.811 *** (0.011)	0.235 *** (0.021)	0.240 *** (0.018)	0.284 *** (0.018)	0.242 *** (0.021)
D (Bulgaria (−1))	−0.036 *** (0.011)	−0.135 *** (0.022)	−0.130 *** (0.019)	−0.144 *** (0.019)	−0.157 *** (0.022)
D (Croatia (−1))	−0.083 *** (0.018)	−0.444 *** (0.037)	0.034 (0.032)	−0.001 (0.032)	−0.011 (0.036)
D (Hungary (−1))	−0.106 *** (0.030)	−0.170 *** (0.060)	−0.589 *** (0.052)	−0.241 *** (0.052)	−0.110 * (0.059)
D (Romania (−1))	−0.635 *** (0.022)	0.235 *** (0.044)	0.222 *** (0.038)	−0.061 (0.038)	0.153 *** (0.043)
D (Slovenia (−1))	0.072 *** (0.022)	0.325 *** (0.043)	0.301 *** (0.037)	0.353 *** (0.038)	−0.098 ** (0.043)
C	0.001 (0.003)	−0.001 (0.006)	0.001 (0.005)	0.001 (0.005)	0.001 (0.006)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The error correction terms indicate that Bulgaria adjusts most rapidly to long-term deviations, with an ECT of −0.811, reflecting its central stabilizing role. In contrast, Croatia, Hungary, Romania, and Slovenia adjust more slowly, with positive ECTs suggesting passive responses to disequilibria. Short-term dynamics reveal notable spillover effects within the block. Bulgaria's prices are significantly influenced by Romania, Hungary, and Croatia, while exhibiting weak self-correction. Croatia shows substantial self-correction but also responds to influences from Romania, Hungary and Bulgaria. Hungary demonstrates robust self-regulation and is particularly sensitive to price changes in Slovenia and Romania. Romania, while strongly negatively influenced by Hungary and Bulgaria, exhibits a significant positive impact from Slovenia, emphasizing its centrality in the block. Slovenia stands out for its self-correcting mechanism and its strong interdependence with Bulgaria and Romania.

Block 3 demonstrates dynamic yet somewhat uneven adjustment patterns. These trends are largely influenced by the region's transitional electricity markets, varying levels of energy infrastructure development, and diverse energy generation portfolios. Bulgaria, as well as Romania and Slovenia (in some periods), are exporters of electricity in the region, driven by their significant low-carbon generation capacities. These countries often play a stabilizing role in the block, providing a steady supply to their neighbors. Also, Slovenia benefits from its strong interconnections with Austria and Italy, which provide access to diverse energy sources. Hungary and Croatia in contrast, rely on electricity imports to meet domestic demand, particularly during periods of high consumption. Hungary, acting as a central hub for electricity flows in Central and Eastern Europe, experiences significant price influences from its interconnected neighbors. Croatia, with its smaller market size, exhibits sensitivity to regional supply fluctuations. Market policies and regulatory environments also shape the block's dynamics. The gradual integration of these countries into the European Union's Internal Electricity Market (IEM) has enhanced cross-border trading and price convergence. However, the region's infrastructure, while improving, still faces challenges related to transmission bottlenecks and limited renewable energy integration compared to Western European countries. These infrastructure limitations, coupled with policy differences in renewable energy support and market liberalization, contribute to variations in adjustment speeds within the block.

Overall, Block 3 exemplifies a highly integrated regional market with Bulgaria and Romania as key drivers of both long-term and short-term dynamics. Slovenia acts as a bridge, connecting various markets through significant mutual influences.

3.3.4. Vector Error Correction Model (VECM) for Block 4

The VECM results for Block 4, encompassing Denmark, Finland, and Sweden, are illustrated in Tables 10 and 11.

Table 10. Cointegration equations for Block 4.

	CointEq1	CointEq2	CointEq3
Denmark (−1)	1.000	−0.222 (0.039)	0.328 (0.051)
Finland (−1)	−4.501 (0.223)	1.000	−1.474 (0.047)
Sweden (−1)	3.054 (0.233)	−0.679 (0.038)	1.000
C	1.871	−0.416	0.613

Table 11. Error correction mechanism for Block 4.

	D (Denmark)	D (Finland)	D (Sweden)
Error Correction Term	−0.007 (0.005)	0.096 *** (0.006)	0.020 *** (0.005)
D (Denmark (−1))	−0.185 *** (0.025)	0.012 (0.029)	0.150 *** (0.026)
D (Finland (−1))	−0.022 (0.026)	−0.155 *** (0.030)	−0.017 (0.027)
D (Sweden (−1))	0.046 (0.033)	0.148 *** (0.039)	−0.173 *** (0.034)
C	0.001 (0.011)	0.001 (0.012)	−0.001 (0.011)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The cointegrating equations highlight significant long-term relationships, where Denmark is positively influenced by Finland (+4.501) and negatively by Sweden (−3.054), reflecting a balancing mechanism between these markets. Finland, normalized in the second equation, is positively influenced by Denmark (+0.222) but negatively by Sweden (−0.679), suggesting that price increases in Sweden tend to counteract price movements in Finland. For Sweden, the long-term dynamics show a negative influence from Denmark (−0.328) and a positive influence from Finland (+1.474), underscoring the role of regional balancing in these integrated markets.

In terms of short-term adjustments, the error correction terms reveal that Finland (0.096) and Sweden (0.020) adjust significantly and positively to deviations from long-term equilibrium, suggesting active responses to restore balance. Denmark (−0.007), however, shows an insignificant and weak adjustment, indicating a limited role in correcting disequilibria in the short term.

The short-term dynamics further emphasize the interconnected nature of these markets. Denmark exhibits strong self-correction (−0.185) and is positively influenced by Sweden (0.046), while Finland exerts a negligible impact. Finland shows significant self-correction (−0.155) and is positively influenced by Sweden (0.148), reflecting the influence of shared infrastructure and market conditions. Sweden also demonstrates self-regulation (−0.173)

and is positively influenced by Denmark (0.150), highlighting Denmark's role in stabilizing Swedish prices.

Overall, Block 4 represents a highly integrated electricity market, underpinned by the robust mechanisms of the Nordic electricity system. Finland plays a leading role in long-term adjustments within this block. This leadership is reflective of Finland's balanced energy mix, which combines nuclear power, hydropower, wind, and biofuels, ensuring stable and predictable electricity generation. The country's strong regulatory framework further enhances its capacity to stabilize the block over the long term. Sweden's influence on short-term dynamics, as highlighted in the analysis, stems from its reliance on hydropower and nuclear energy, which provide flexibility to quickly respond to fluctuations in demand and supply. The country's ability to modulate hydropower production enables it to act as a stabilizing force, particularly during periods of volatility caused by Denmark's wind energy variability. Denmark, while exhibiting strong self-regulation, plays a less dominant role in driving adjustments within the block. Its significant reliance on wind power, which is inherently intermittent, leads to more localized price adjustments that are less influential in shaping the broader market dynamics. However, Denmark's proactive policies on renewable energy integration and its pioneering role in cross-border energy trading ensure its continued contribution to the block's overall stability. The advanced interconnection infrastructure within Block 4, facilitated by Nord Pool, supports the high level of integration and price convergence observed in this market.

3.3.5. Vector Error Correction Model (VECM) for Block 5

The VECM results for Block 5, comprising Estonia, Latvia, and Lithuania, are indicated in Tables 12 and 13.

Table 12. Cointegration equations for Block 5.

	CointEq1	**CointEq2**	**CointEq3**
Estonia (−1)	1.000	−0.208 (0.012)	0.258 (0.016)
Latvia (−1)	−4.815 (0.232)	1.000	−1.242 (0.015)
Lithuania (−1)	3.878 (0.231)	−0.805 (0.012)	1.000
C	−0.249	0.052	−0.064

Table 13. Error correction mechanism for Block 5.

	D (Estonia)	**D (Latvia)**	**D (Lithuania)**
Error Correction Term	−0.274 *** (0.045)	−0.068 (0.045)	−0.125 *** (0.045)
D (Estonia (−1))	0.021 (0.067)	0.139 ** (0.067)	0.176 *** (0.067)
D (Latvia (−1))	−0.772 *** (0.249)	−0.567 ** (0.249)	−0.399 (0.249)
D (Lithuania (−1))	0.598 *** (0.232)	0.270 (0.232)	0.071 (0.232)
C	0.001 (0.007)	0.001 (0.007)	0.001 (0.007)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The cointegrating equations highlight significant long-term relationships, with Estonia as the reference in the first equation. Latvia (+4.815) exerts a dominant positive influence

on Estonia, indicating that price increases in Latvia drive corresponding increases in Estonia. Lithuania (−3.878), on the other hand, has a negative influence, suggesting a balancing effect between these two markets. In the second equation, normalized on Latvia, Estonia (+0.208), and Lithuania (+0.805), both exhibit positive impacts, emphasizing their mutual alignment with Latvia. In the third equation, Lithuania is normalized, and its dynamics are shaped by positive effects from Latvia (+1.242) and negative contributions from Estonia (−0.258).

The error correction terms illustrate the speed and significance of adjustments to deviations from the long-term equilibrium. Estonia (−0.274) and Lithuania (−0.125) adjust significantly, indicating their active role in stabilizing the market. Latvia (−0.068) shows weaker and insignificant adjustment, suggesting a more passive approach to restoring equilibrium.

Short-term dynamics emphasize the influence of past price changes within the block. Estonia's prices are positively influenced by Lithuania (0.598), reflecting strong integration between these two markets. Latvia negatively affects Estonia (−0.772), suggesting balancing dynamics. Latvia, in turn, is positively influenced by Estonia (0.139), exhibiting negative self-correction (−0.567), while Lithuania has no significant short-term effect. Lithuania exhibits positive self-correction (0.071) and is influenced significantly by Estonia (0.176).

Overall, Block 5 reflects the tightly integrated nature of the Baltic electricity market, shaped by shared energy infrastructure, interdependencies, and policy initiatives aimed at reducing dependence on external energy sources. Estonia emerges as a key player in stabilizing long-term imbalances within the block. This influence can be attributed to its significant reliance on domestically available oil shale for electricity generation, which provides stability in supply and predictability in market behavior. Estonia's strategic role in the Baltic Electricity Market Interconnection Plan further underscores its importance in ensuring regional stability. Latvia, while contributing meaningfully to long-term dynamics, exhibits a slower adjustment to market imbalances in the short term. This slower response is likely linked to its reliance on hydropower, which, while flexible, is heavily influenced by seasonal variations in water availability. Additionally, Latvia's intermediate position in the Baltic electricity grid allows it to act as a conduit for power flows between Estonia and Lithuania, highlighting its structural role rather than a leading influence in market adjustments. Lithuania, described in the analysis as the most balanced market, demonstrates reciprocal interactions with both Estonia and Latvia. This balance stems from Lithuania's diversified energy mix and its proactive efforts to integrate renewable energy sources, such as hydropower, wind and solar, while gradually reducing reliance on imported electricity from third countries. The synchronization of the Baltic electricity grids with the European Continental Network, expected to be fully operational by 2025, further enhances Lithuania's ability to dynamically interact with its neighbors, fostering greater integration and market stability. The integration and coordination within Block 5 are further reinforced by shared participation in initiatives like Nord Pool and the Baltic Energy Market Interconnection Plan, which aim to harmonize market operations and promote energy security.

3.3.6. Vector Error Correction Model (VECM) for Block 6

The VECM results for Block 6, comprising Greece, Italy, and Poland, are revealed in Tables 14 and 15.

The cointegrating equations reveal strong long-term relationships, with Greece normalized in the first equation. Italy (+4.849) exerts a dominant positive influence on Greece, indicating that price increases in Italy are associated with increases in Greece. Poland (−6.171) has a strong negative impact, suggesting that price movements in Poland counterbalance those in Greece. In the second equation, normalized on Italy, Greece (+0.206), and Poland (+1.273) both positively influence Italy, emphasizing their alignment with Italian

price movements. In the third equation, Poland is normalized, with Italy (+0.786) and Greece (−0.162) exerting smaller balancing effects.

Table 14. Cointegration equations for Block 6.

	CointEq1	CointEq2	CointEq3
Greece (−1)	1.000	−0.206 (0.054)	0.162 (0.075)
Italy (−1)	−4.849 (0.249)	1.000	−0.786 (0.068)
Poland (−1)	6.171 (0.339)	−1.273 (0.066)	1.000
C	−9.335	1.925	−1.513

Table 15. Error correction mechanism for Block 6.

	D (Greece)	D (Italy)	D (Poland)
Error Correction Term	−0.010 *** (0.002)	−0.01 (0.002)	−0.047 *** (0.003)
D (Greece (−1))	−0.253 *** (0.023)	0.020 (0.020)	0.065 ** (0.031)
D (Italy (−1))	0.170 *** (0.028)	−0.171 *** (0.024)	0.007 (0.038)
D (Poland (−1))	0.002 (0.017)	−0.006 (0.014)	−0.106 *** (0.022)
C	0.001 (0.003)	0.001 (0.003)	0.001 (0.004)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The error correction terms indicate the speed and significance of adjustments to deviations from the long-term equilibrium. Greece (−0.010) and Poland (−0.047) adjust significantly, with Poland demonstrating the strongest adjustment among the three countries. Italy (−0.01), while showing adjustment, exhibits a smaller and less significant role in correcting disequilibria.

Short-term dynamics highlight the spillover effects and interdependencies within the block. Greece's prices are significantly influenced by Italy (0.170), indicating a robust connection between these markets. Poland (0.002) also positively influences Greece, though to a lesser extent. Italy's prices are weakly negatively influenced by Greece (−0.020), reflecting a balancing effect, and exhibit strong self-correction (−0.171). Poland, while self-regulating (−0.106), shows limited influence from Greece and Italy in the short term.

Overall, Block 6 highlights the interconnected dynamics of Greece, Italy, and Poland, driven by their diverse energy infrastructures and distinct market policies. Italy plays a pivotal role as a key driver of market dynamics in this block, exerting significant influence on both Greece and Poland. This dominance is largely attributable to Italy's position as one of the largest electricity markets in Europe, with a highly diversified energy mix that includes substantial contributions from natural gas, renewables, and imports through cross-border interconnections. Italy's extensive participation in the European electricity market, supported by its robust transmission network, allows it to shape both long-term and short-term price adjustments within the block. Greece, while steadily adjusting to long-term equilibrium, exhibits significant short-term dependencies on Italy and Poland. This can be explained by Greece's reliance on natural gas imports, which are often subject to price fluctuations influenced by Italy's energy market dynamics. Additionally, Greece's

increasing share of renewable energy, particularly solar and wind, introduces variability that strengthens its dependency on neighboring countries to maintain market balance. Poland demonstrates robust self-regulation and plays a critical role in stabilizing long-term dynamics within the block. This is reflective of Poland's reliance on coal-fired power plants, which provide a stable but environmentally challenging energy base. At the same time, Poland's growing investments in renewable energy and interconnection projects with neighboring countries have enhanced its ability to balance the market dynamics within the block. Poland's strategic location and its role as a transit hub for electricity flows further underscore its importance in maintaining equilibrium in Block 6.

3.3.7. Vector Error Correction Model (VECM) for Block 7

The VECM results for Block 7, comprising Portugal and Spain, are highlighted in Tables 16 and 17.

Table 16. Cointegration equations for Block 7.

	CointEq1	**CointEq2**
Portugal (−1)	1.000	−0.991 (0.002)
Spain (−1)	−1.010 (0.002)	1.000
C	0.035	−0.035

Table 17. Error correction mechanism for Block 7.

	D (Portugal)	**D (Spain)**
Error Correction Term	−0.391 (0.162)	0.354 ** (0.160)
D (Portugal (−1))	0.199 (0.129)	0.260 ** (0.127)
D (Spain (−1))	−0.191 (0.130)	−0.238 (0.128)
C	0.001 (0.007)	0.001 (0.007)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The cointegrating equations highlight the symmetric long-term equilibrium between them. In the first equation, normalized on Portugal, Spain (+1.010) exerts a strong positive influence, indicating that price increases in Spain lead to price increases in Portugal over the long term. Similarly, in the second equation, normalized on Spain, Portugal (+0.991) has a significant and positive influence on Spain. This symmetry underscores the mutual dependence of these markets, driven by their shared infrastructure and similar market conditions.

The error correction terms (ECTs) reveal how the markets adjust to deviations from the long-term equilibrium. Spain (0.354) adjusts significantly, reflecting its active role in restoring balance within the block. Portugal (0.391) is adjusting at a less significant rate.

Short-term dynamics further emphasize the interconnected nature of the two markets. Portugal's prices are positively influenced by its own lagged values (0.199) and negatively influenced, though not significantly, by Spain (−0.191). Conversely, Spain exhibits self-correction and a positive influence from Portugal (0.260). These dynamics suggest that while both countries influence each other, Spain has a slightly more dominant short-term role.

Overall, Block 7 highlights the strong integration between the electricity markets of Portugal and Spain, reflecting their shared geographical, infrastructural, and regulatory

frameworks. The symmetric long-term relationships observed in this block underscore the deep interconnection between these two countries, facilitated by the Iberian Electricity Market (MIBEL). Established to integrate their electricity systems, MIBEL has enabled efficient cross-border electricity trading, fostering a unified market structure that supports balanced price adjustments. Spain emerges as the more active stabilizer within the block, efficiently responding to long-term imbalances. This is largely due to Spain's significant market size and diversified energy mix, which includes a substantial share of renewable energy such as wind and solar power. Spain's advanced transmission and distribution network, along with its leading role in renewable energy integration, enables it to absorb and adjust to fluctuations more effectively than Portugal. Moreover, Spain's role as a regional energy hub further strengthens its capacity to stabilize the market. Portugal, while closely interconnected with Spain, demonstrates a greater dependency on its neighbor for short-term adjustments. This is indicative of Portugal's smaller market size and its reliance on electricity imports from Spain during periods of heightened demand or renewable energy variability. Portugal's growing renewable energy sector, particularly hydropower, contributes to the block's overall integration but also introduces seasonal dependencies that Spain helps to balance. The dynamics within Block 7 exemplify a well-integrated regional market, where Spain's robust market infrastructure and active stabilization efforts complement Portugal's growing renewable capabilities.

3.4. Block VECM Between Clusters/Blocks

The VECM results for the inter-block dynamics, among the seven aggregated electricity market blocks in the European Union, are highlighted in Tables 18 and 19.

Table 18. Cointegration equations for all blocks.

	CointEq1	CointEq2	CointEq3	CointEq4	CointEq5	CointEq6	CointEq7
Block 1 (−1)	1.000	−3.086 (0.102)	−2.889 (0.107)	94.283 (3.589)	449.823 (16.521)	−2.044 (0.077)	8.676 (0.322)
Block 2 (−1)	−0.324 (0.030)	1.000	0.936 (0.098)	−30.555 (3.048)	−145.777 (15.248)	0.663 (0.070)	−2.812 (0.261)
Block 3 (−1)	−0.346 (0.050)	1.068 (0.157)	1.000	−32.639 (4.748)	−155.721 (22.971)	0.708 (0.077)	−3.004 (0.443)
Block 4 (−1)	0.011 (0.018)	−0.033 (0.052)	−0.031 (0.049)	1.000	4.771 (7.199)	−0.022 (0.036)	0.092 (0.152)
Block 5 (−1)	0.002 (0.033)	−0.007 (0.104)	−0.006 (0.097)	0.210 (2.894)	1.000	−0.005 (0.067)	0.019 (0.294)
Block 6 (−1)	−0.489 (0.055)	1.509 (0.172)	1.413 (0.118)	46.117 (5.254)	−220.024 (24.331)	1.000	−4.244 (0.484)
Block 7 (−1)	0.115 (0.016)	−0.356 (0.045)	−0.333 (0.048)	10.867 (1.559)	51.846 (7.465)	−0.236 (0.034)	1.000
C	0.345	−1.066	−0.998	32.566	155.373	−0.706	2.997

The cointegration equations highlight Block 1 as a central hub, showing substantial positive long-term relationships with Blocks 2 (+0.324) and 3 (+0.346), as well as strong influence from Block 6 (+0.489), while being counterbalanced by Block 7 (−0.115). Block 2 aligns closely with Block 3 (−1.068), indicating a mutual dependence, and receives notable influence from Blocks 1 and 6. Block 3, in turn, demonstrates reciprocal relationships with Blocks 1 and 2 and is driven, also, by Block 6 (−1.413), with counterbalancing effects from Block 7 (+0.333). Blocks 4 and 5 play a more independent role, contributing minimally to the long-term dynamics, while Block 7 acts as a balancing force across the system.

Table 19. Error correction mechanism for all blocks.

	D (B1)	D (B2)	D (B3)	D (B4)	D (B5)	D (B6)	D (B7)
Error Correction Term	−0.741 *** (0.031)	−0.313 *** (0.029)	−0.079 *** (0.017)	−0.374 *** (0.039)	−0.280 *** (0.028)	−0.062 *** (0.011)	−0.182 *** (0.029)
D (B1 (−1))	0.054 * (0.031)	0.199 *** (0.028)	0.068 *** (0.016)	0.206 *** (0.038)	0.170 *** (0.028)	0.044 *** (0.010)	0.114 *** (0.028)
D (B2 (−1))	0.080 *** (0.032)	−0.266 *** (0.029)	0.071 *** (0.017)	0.098 *** (0.039)	0.087 *** (0.029)	0.042 *** (0.011)	−0.040 (0.029)
D (B3 (−1))	−0.303 *** (0.058)	−0.157 *** (0.053)	−0.433 *** (0.031)	−0.164 ** (0.072)	−0.079 (0.052)	−0.030 (0.020)	−0.151 *** (0.053)
D (B4 (−1))	0.065 *** (0.021)	0.041 ** (0.019)	0.040 *** (0.011)	−0.089 *** (0.025)	0.158 *** (0.019)	0.046 *** (0.007)	0.046 ** (0.019)
D (B5 (−1))	0.010 (0.031)	0.001 (0.028)	−0.014 (0.016)	0.016 (0.038)	−0.320 *** (0.028)	−0.001 (0.010)	−0.010 (0.028)
D (B6 (−1))	−0.117 (0.088)	−0.166 ** (0.081)	0.025 (0.047)	−0.476 *** (0.109)	−0.285 *** (0.080)	−0.338 *** (0.030)	−0.145 * (0.081)
D (B7 (−1))	0.112 *** (0.023)	0.158 *** (0.022)	0.063 *** (0.013)	0.068 ** (0.029)	0.081 *** (0.021)	0.036 *** (0.008)	0.055 ** (0.022)
C	0.001 (0.007)	−0.001 (0.007)	0.001 (0.004)	0.001 (0.009)	0.001 (0.007)	0.001 (0.003)	0.001 (0.007)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The error correction terms indicate how quickly each block adjusts to long-term disequilibria. Blocks 1 (−0.741), 4 (0.374), and 2 (−0.313) exhibit the fastest adjustments, underscoring their stabilizing roles, while Blocks 6 (−0.062) and 3 (−0.079) adjust more gradually, reflecting their supporting roles in equilibrium restoration. Short-term dynamics reveal significant spillovers, with Blocks 3, 6, and 5 influenced by their own past values. Blocks 6, 3, and 1 emerge as key transmitters of short-term influences.

Overall, Blocks 1, 2, and 3 dominate both long-term and short-term interactions, serving as central players in the interconnected system. Block 7 bridges the dynamics across regions, while Blocks 4 and 5 maintain relative independence. Block 6, despite slower adjustments, exerts balancing effects, highlighting the complex interdependence of these aggregated electricity markets.

4. Conclusions

This study investigates the price behavior and interconnectedness of electricity markets within the European Union (EU) using the Vector Error Correction Model (VECM) to understand both long-term equilibrium relationships and short-term dynamics. Daily wholesale day-ahead electricity prices from 24 EU countries, spanning October 2017 to September 2024, provide a comprehensive foundation for evaluating regional electricity price trends and interdependencies.

The findings highlight a high degree of integration among EU electricity markets, with distinct regional dynamics observed across the identified clusters. Germany, for instance, emerges as a pivotal player in Block 1, influencing both Austria and Czechia significantly. Similarly, France and Belgium play central roles in driving dynamics in Block 2, while countries such as Romania and Bulgaria demonstrate strong interconnections in Block 3. For Blocks 4 and 5, Scandinavian and Baltic markets exhibit unique balancing mechanisms, influenced heavily by shared infrastructure and market policies. The aggregated inter-block VECM analysis further highlights the centrality of Blocks 1 and 3. Block 1 demonstrates strong interconnections with Blocks 2 and 3, underscoring its role as a stabilizing hub in

the EU's electricity market. Similarly, Block 3 exhibits reciprocal relationships with Blocks 1 and 2, reflecting its importance in bridging Eastern and Western European markets. Blocks 4 and 5 maintain more independent dynamics, while Block 7 acts as a critical intermediary, transmitting influences across regions.

These results highlight the EU's progress toward market integration, with central players driving stability and cohesion across the electricity markets. The clustering approach reveals regional patterns that align with geographical, infrastructural, and policy similarities, underscoring the importance of targeted interventions to address disparities and inefficiencies. The inter-block analysis confirms the strategic importance of central hubs and the need for coordinated efforts to enhance resilience and adaptability in the interconnected grid.

The insights from this study are of considerable interest to multiple stakeholders. The results provide valuable guidance for enhancing cross-border electricity market policies and addressing market inefficiencies. Policymakers can use these findings to prioritize investment in infrastructure that strengthens market integration, particularly in regions where asymmetrical dependencies are observed. Also, the market participants (energy producers, distributors, and traders) can leverage the findings to develop informed pricing and trading strategies. By understanding the drivers of short-term and long-term price adjustments, these participants can optimize operations and reduce risks. Moreover, for researchers and analysts the methodology and insights contribute to the broader field of energy economics, offering a replicable framework for analyzing interconnected markets.

While this study provides comprehensive insights, there are limitations that warrant further exploration. Key among these is the reliance on the daily wholesale day-ahead electricity prices (which may not capture all the nuances of the relationship between national electricity markets), simplified assumptions (the models assume uniform behavior within clusters), and the exclusion of some external factors (factors such as natural gas prices, carbon allowances, and seasonal demand variations were not explicitly modeled but could have significant implications for electricity price dynamics). Future research could address these limitations by exploring the interplay between electricity prices and other macroeconomic or environmental factors. Moreover, future research can involve conducting a comprehensive sensitivity analysis, applied to the model, under different scenarios. Such scenarios could include varying energy prices or fluctuations in renewable energy production levels. This type of analysis would provide deeper insights into the dynamics and resilience of European electricity markets under diverse economic and policy conditions.

Also, to address the limitations of traditional models and incorporate recent advancements, future research could benefit from a hybrid approach that integrates econometric techniques with machine learning methodologies. For example, combining VECM with reinforcement learning could enhance the adaptability of equilibrium-based models to dynamic market conditions. Leveraging time series GANs alongside VECM could improve the robustness of long-term forecasts by accounting for non-linearities and structural breaks. Utilizing LLMs to preprocess unstructured data, such as policy documents and news articles, could enrich traditional models with contextual insights.

The findings lead to several recommendations. Firstly, the policies aimed at fostering cross-border electricity trading and enhancing grid connectivity should focus on central hubs (such as Germany and France). These markets exhibit strong spillover effects and can serve as anchors for regional stability. Secondly, countries or clusters with slower adjustment mechanisms could benefit from targeted policies and investments to enhance their responsiveness to market changes. Addressing these bottlenecks would improve overall system efficiency. Finally, given the importance of infrastructure and market

interconnections, a coordinated EU-wide strategy for integrating renewable energy sources into the grid is essential for maintaining equilibrium and addressing regional disparities.

The literature on electricity price dynamics has evolved significantly, with traditional econometric models providing a robust foundation for understanding market integration and advanced AI techniques offering new possibilities for innovation. While this study emphasizes the utility of VECM for analyzing European electricity markets, future research should explore the integration of traditional and advanced methodologies to achieve a more comprehensive understanding of market behavior.

The authors consider that the study contributes to a deeper understanding of how regional power interdependence impacts local electricity markets constrained by natural monopolies and physical infrastructure. By identifying the roles of key market blocks in stabilizing the interconnected system, the research offers actionable insights for policymakers to optimize cross-border cooperation and infrastructure investments. These findings emphasize the potential of regional integration to address local market constraints and improve energy access and reliability.

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